Turtle Games Technical Report

By Luke Whitmore

Word count 1,030 (excluding titles and appendices)

Introduction

Turtle Games is a game manufacturer and retailer with a global customer base. They manufacture and sell their own products, and source and sell other companies' products. Its product range includes books, board games, video games, and toys.

They want to better understand what drives customer loyalty, how customers can be segmented for more effective targeted marketing, whether customer reviews and summaries can provide meaningful insights, and whether predictive models can accurately predict customer loyalty point values.

These insights will help them to better understand its customers, improve the targeting of marketing efforts, enhance customer engagement, and ultimately, increase sales revenue.

Analytical Approach

Data Collection and Preparation:

The required Python libraries, including pandas, numpy, sklearn, matplotlib, seaborn, NLTK, TextBlob, and WordCloud, were imported into a Jupyter Notebook. The turtle_reviews CSV file was then loaded into a DataFrame called 'reviews' for analysis.

The dataset was inspected using the head, dtypes, and info functions, which identified no null values.

The stakeholder advised that the 'language' and 'platform' variables were unnecessary, so these columns were removed. The 'remuneration (£)' and 'spending_score (1-100)' columns were renamed to 'remuneration' and 'spending_score', respectively.

The DataFrame 'reviews_nec_col' was then saved as a CSV file for use in exploratory data analysis in R. The tidyverse, dplyr, ggplot2, and moments libraries were used in R.

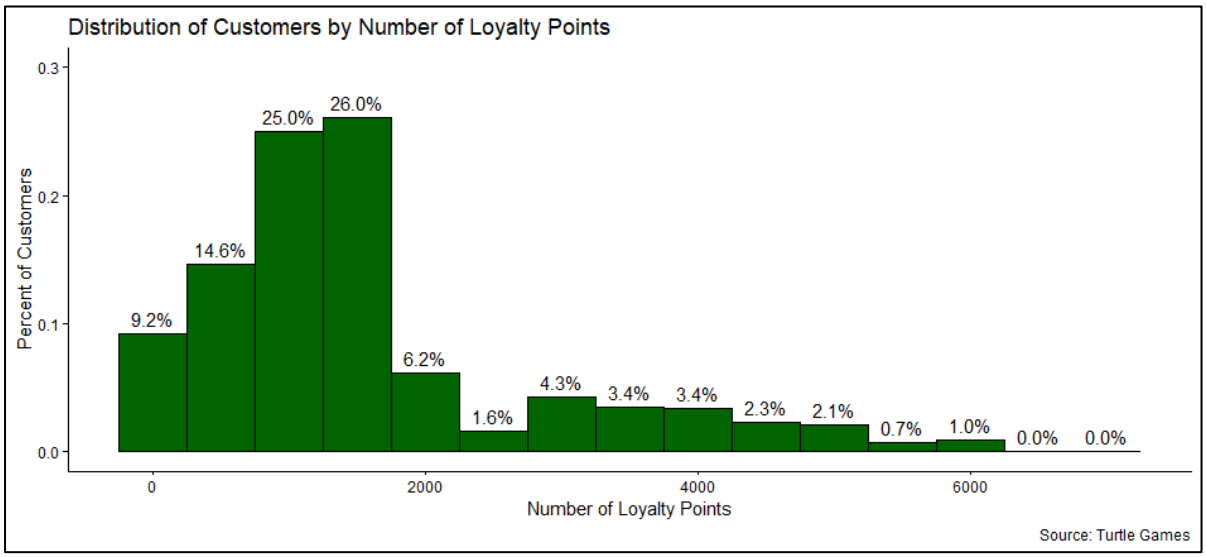
Data Exploration:

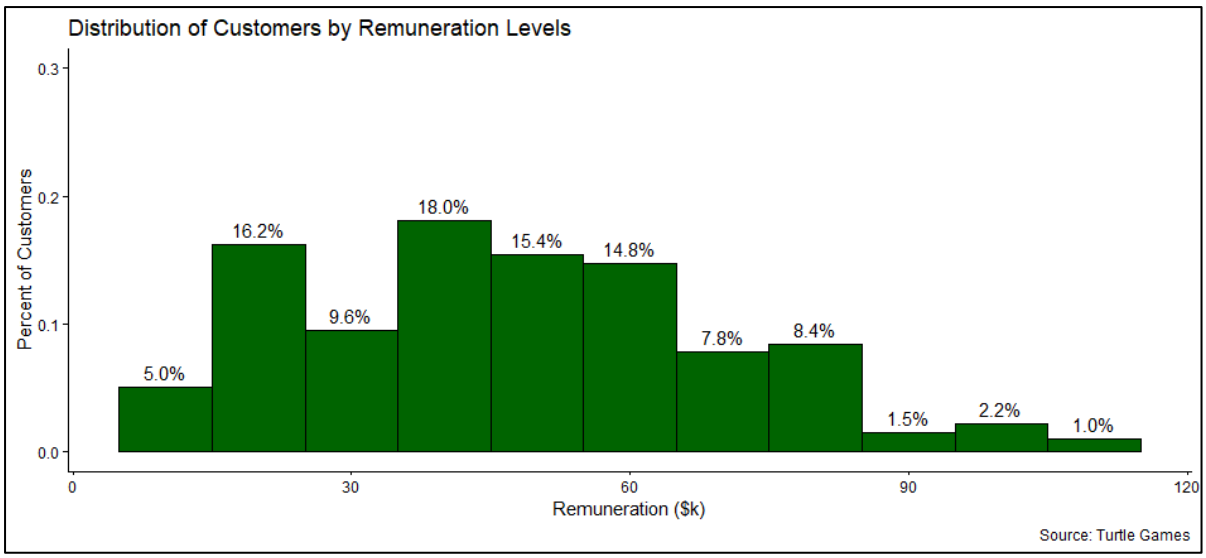
The describe function was used to produce summary statistics of the dataset. These indicated low variability within each column but a high likelihood of outliers in the loyalty_points, remuneration, and spending_score columns.

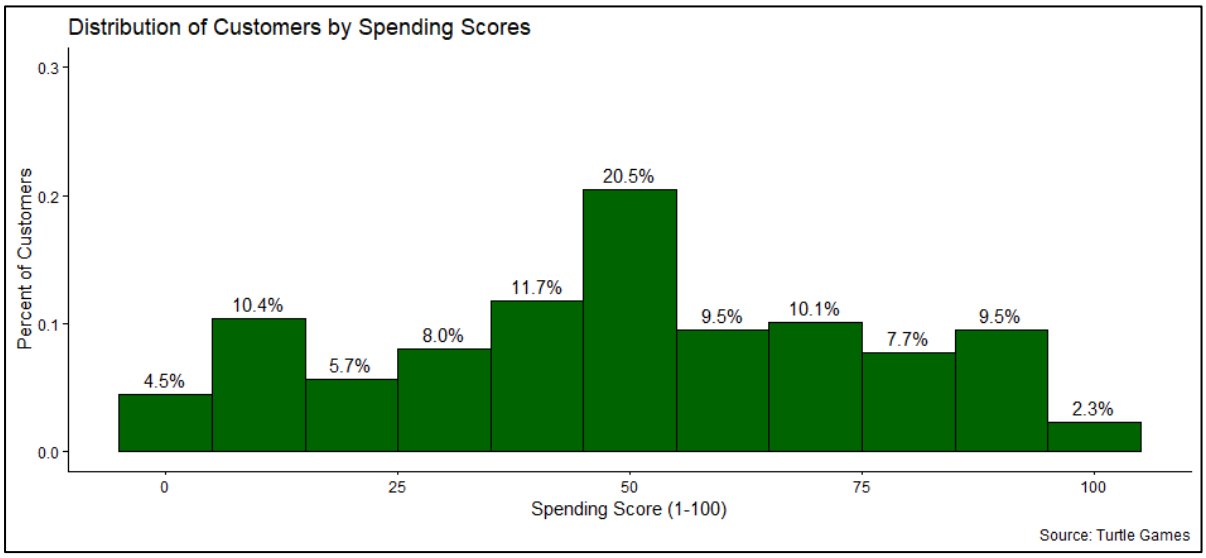
[6]:

	age	remuneration (k£)	spending_score (1-100)	loyalty_points	product
count	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000
mean	39.495000	48.079060	50.000000	1578.032000	4320.521500
std	13.573212	23.123984	26.094702	1283.239705	3148.938839
min	17.000000	12.300000	1.000000	25.000000	107.000000
25%	29.000000	30.340000	32.000000	772.000000	1589.250000
50%	38.000000	47.150000	50.000000	1276.000000	3624.000000
75%	49.000000	63.960000	73.000000	1751.250000	6654.000000
max	72.000000	112.340000	99.000000	6847.000000	11086.000000

Histograms were produced in R to show the distributions of loyalty point, remuneration, and spending score values. This provided early, high-level information on different customer characteristics.





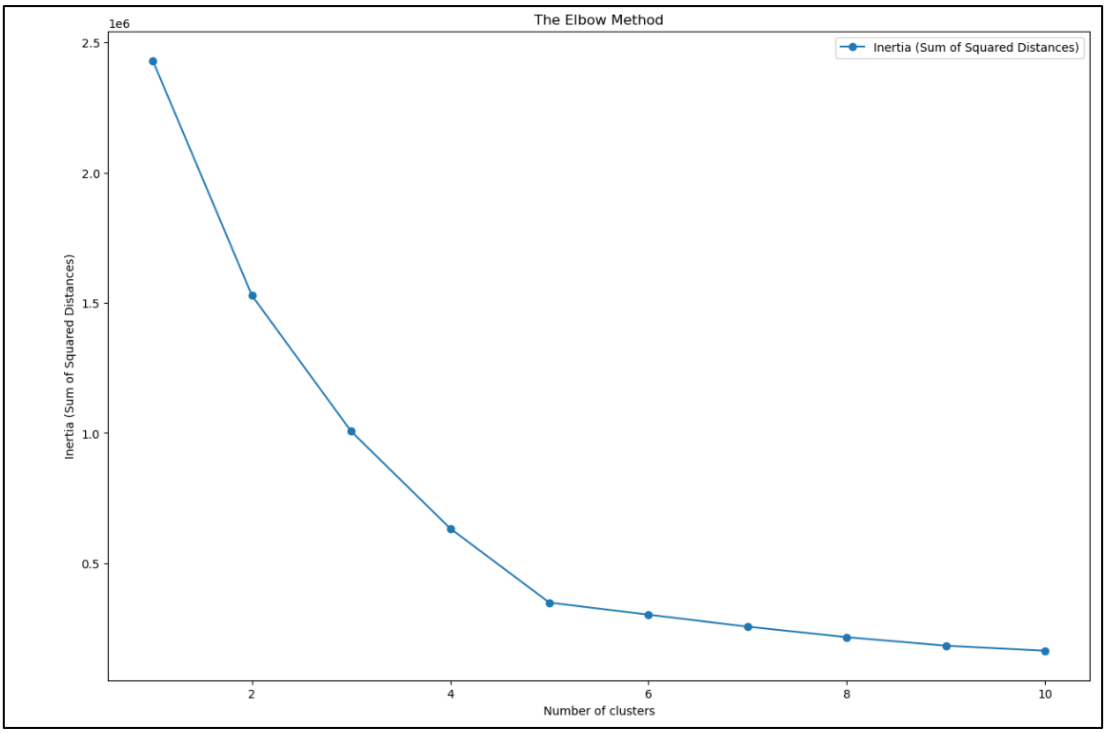


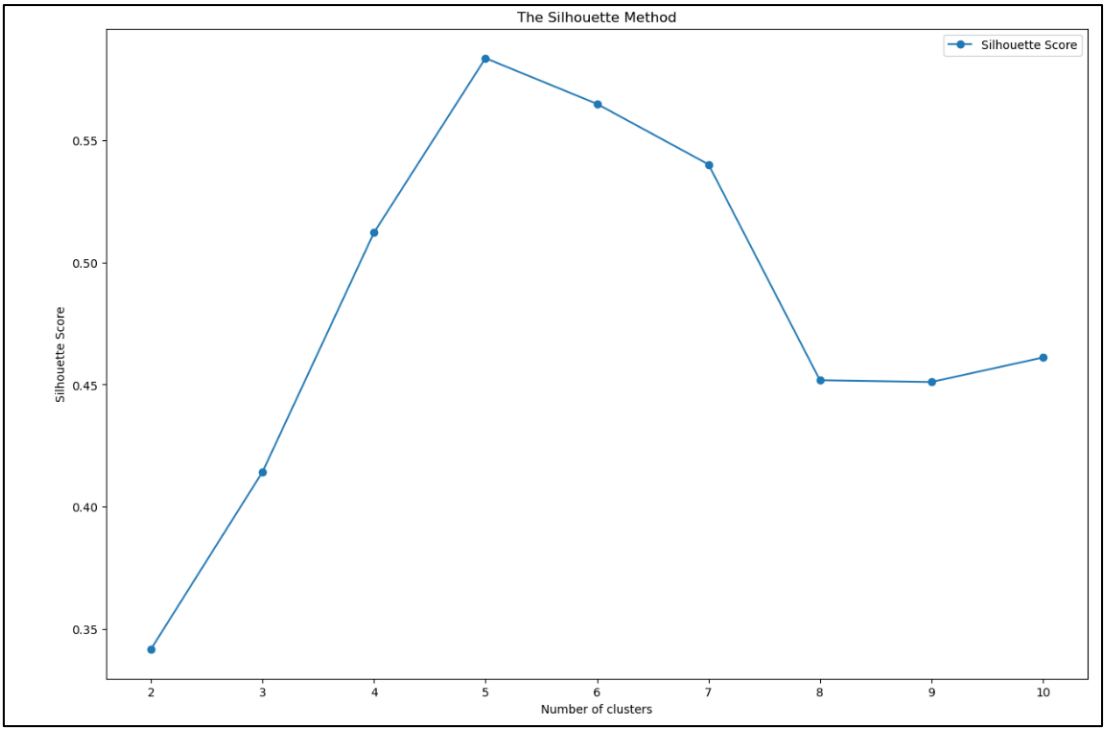
Scatterplots were produced in both Python and R to show the relationships between the dependent variable (loyalty points) and some independent variables (remuneration, spending score, and age). This provided early, high-level information on which variables had the greatest potential to predict loyalty point values (see visuals and insights below).

Predictive Modelling:

Linear regression was conducted in Python using sklearn to quantify relationships between loyalty points and selected explanatory variables. Decision tree analysis was conducted in Python using sklearn to identify the most important variables influencing loyalty point accumulation. Both these methods were useful in determining the predictive power of different independent variables.

K-means clustering was conducted in Python using sklearn, which helped to identify naturally occurring customer segments. The elbow and silhouette methods were used to determine the most appropriate number of clusters.





Natural language processing was conducted in Python using NLTK, TextBlob, and WordCloud. This involved text cleaning, tokenisation, frequency analysis, and sentiment analysis, to assess whether customer reviews and summaries could provide useful business insights.

Multiple linear regression was conducted in R to assess whether loyalty points could be predicted accurately.

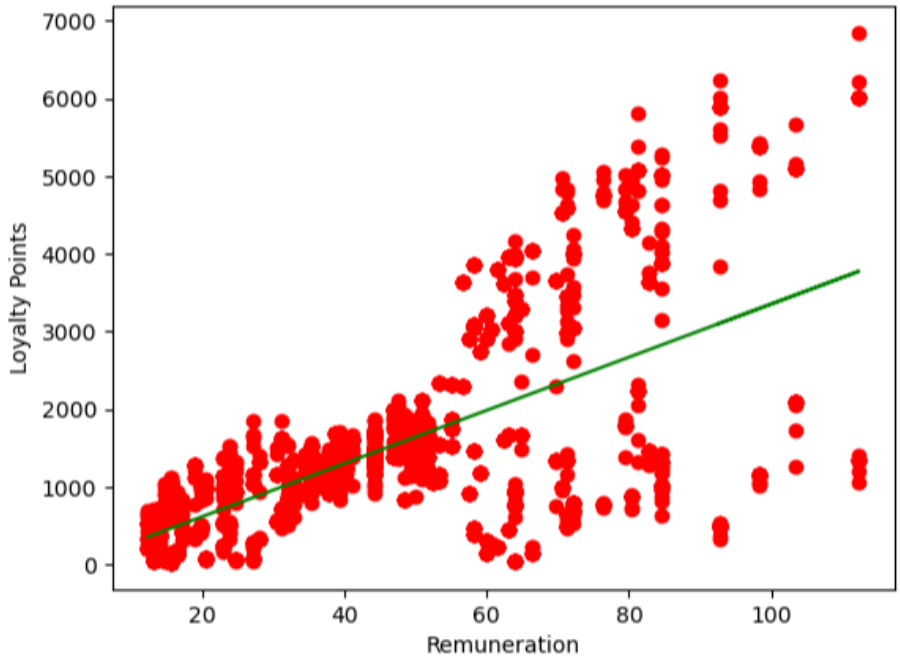
Reproducibility and Documentation:

Markdown was used in the Python and R analyses to explain code, outputs and observations. Analyses followed a structured workflow from data preparation through to interpretation. Findings were validated in R to improve reliability and reproducibility.

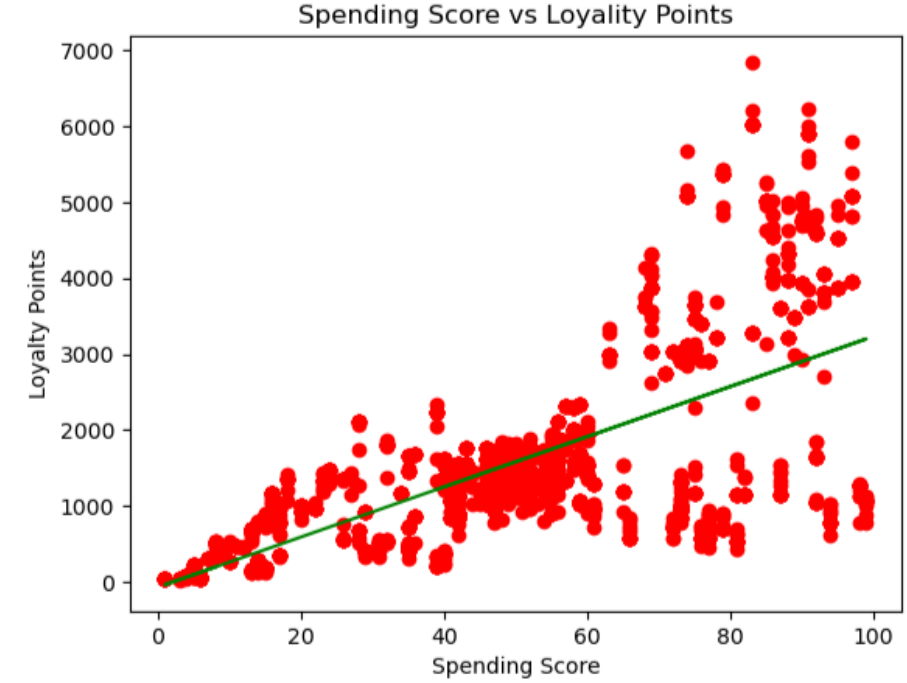
Visualisations and Insights

Q1. What drives loyalty point accumulation?

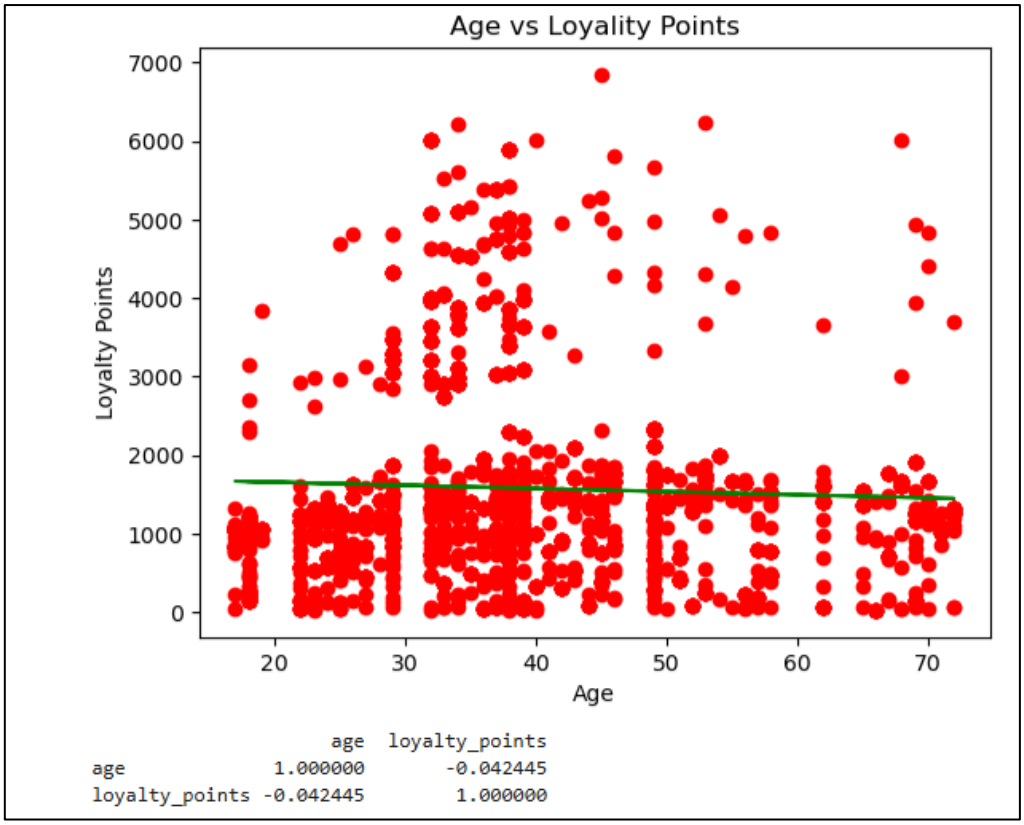
Visualisations from the linear regression and decision tree analyses show that remuneration and spending score are the strongest predictors of loyalty point accumulation.



	remuneration	loyalty_points
remuneration	1.000000	0.616065
loyalty_points	0.616065	1.000000

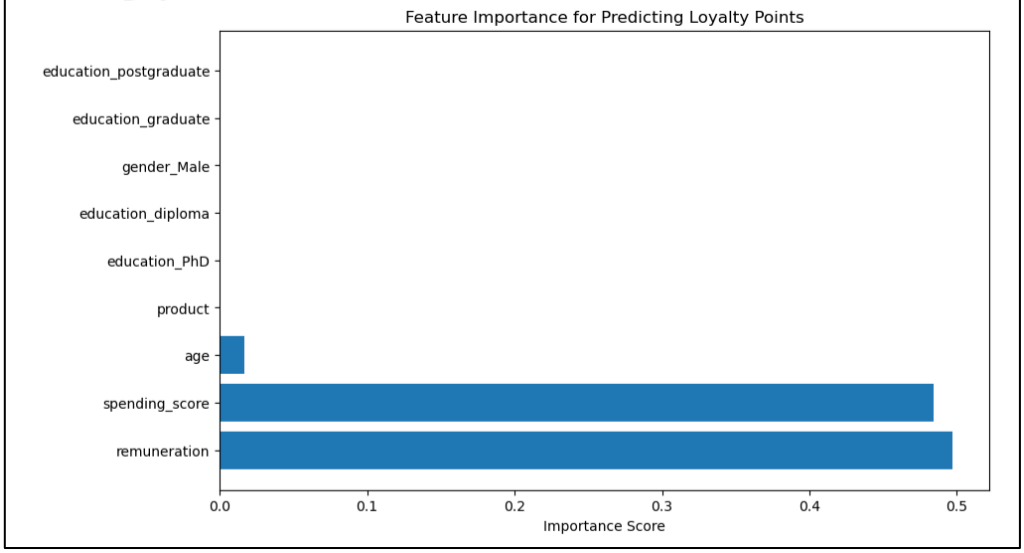


	spending_score	loyalty_points
spending_score	1.00000	0.67231
loyalty_points	0.67231	1.00000



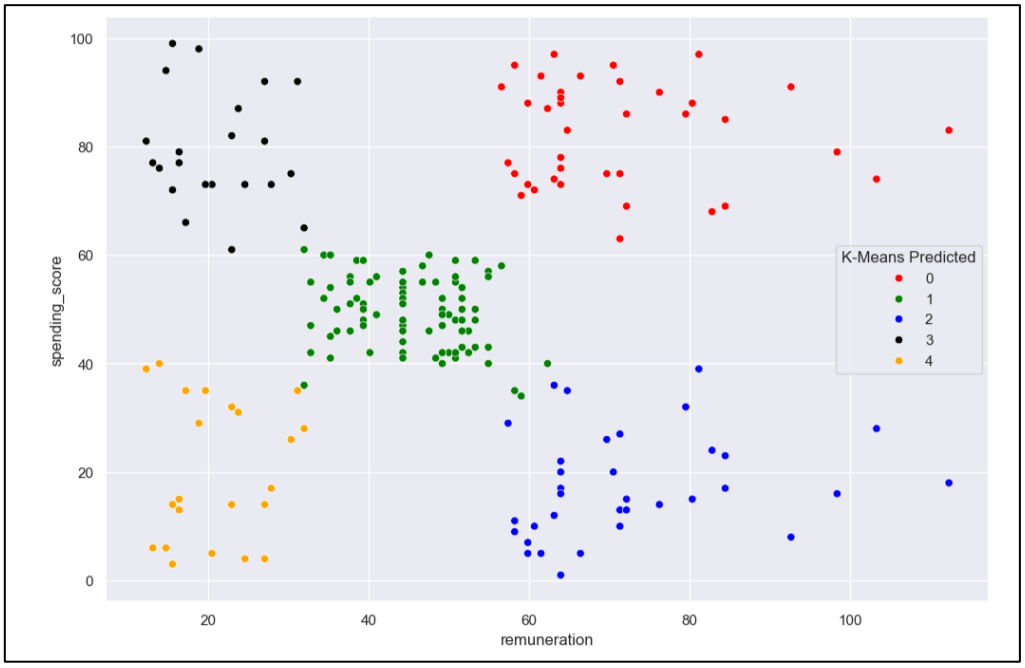
	age	loyalty_points
age	1.000000	-0.042445
loyalty_points	-0.042445	1.000000

	Feature	Importance
1	remuneration	0.497246
2	spending_score	0.483997
0	age	0.016846
3	product	0.000844
5	education_PhD	0.000324
6	education_diploma	0.000237
4	gender_Male	0.000226
7	education_graduate	0.000202
8	education_postgraduate	0.000078



Q2. How can customers be segmented?

The clustering visualisations identified five customer segments, three of which demonstrated greater importance due to their size, value, or growth potential.

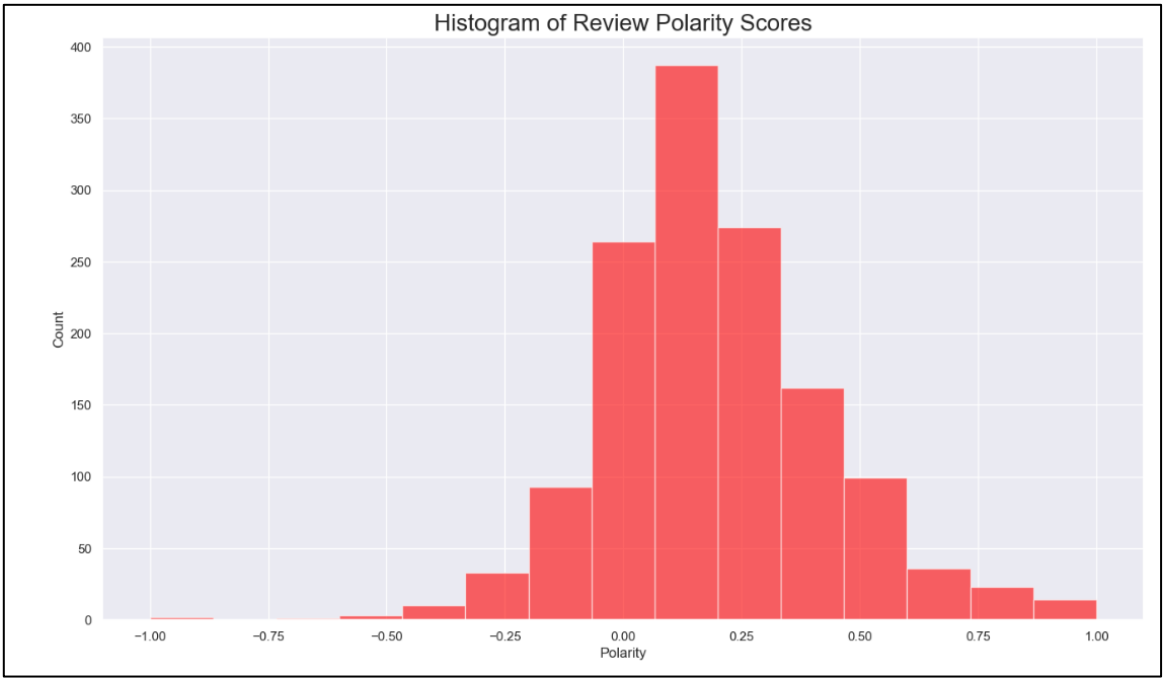


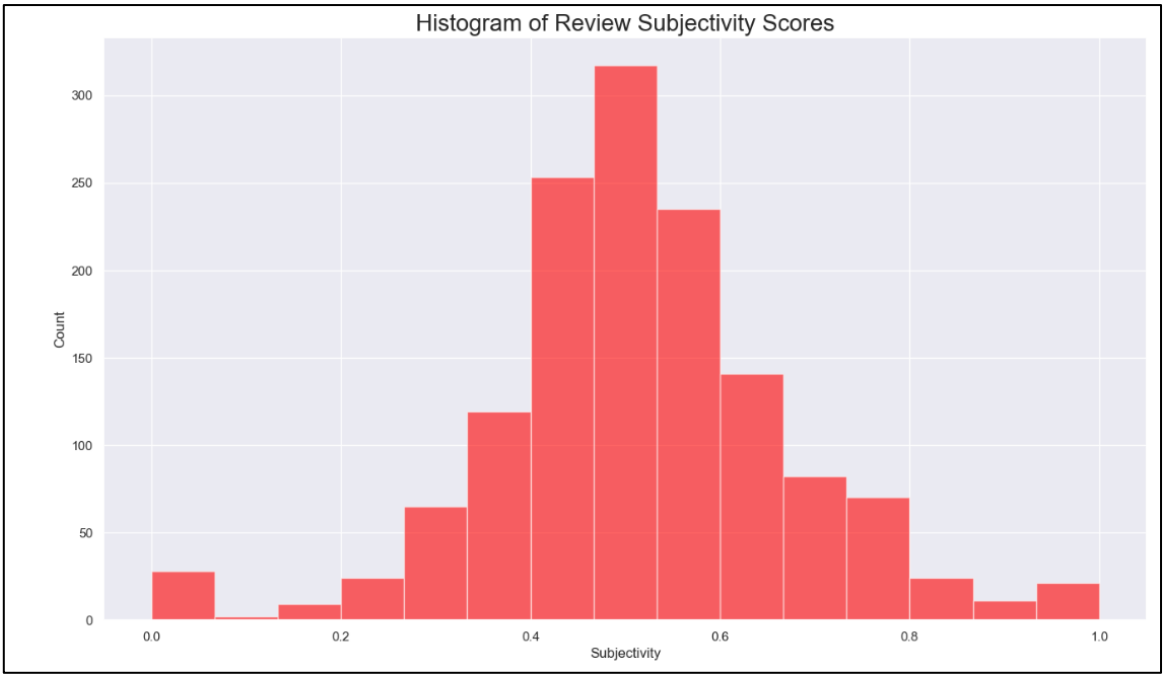
Cluster No.	Cluster Colour	Cluster Name	No. of Customers	Business Priority
2	Blue	High earners and low spenders	330	Highest growth opportunity
0	Red	High earners and high spenders	356	Highest current value
1	Green	Moderate earners and moderate spenders	774	Largest segment
3	Black	Low earners and high spenders	269	Lower priority
4	Orange	Low earners and low spenders	271	Lowest priority

Q3. Can reviews provide useful insights?

The word cloud and sentiment visualisations suggested that customer sentiment was generally neutral to positive, with customers frequently discussing enjoyment, gameplay, and positive product experiences.

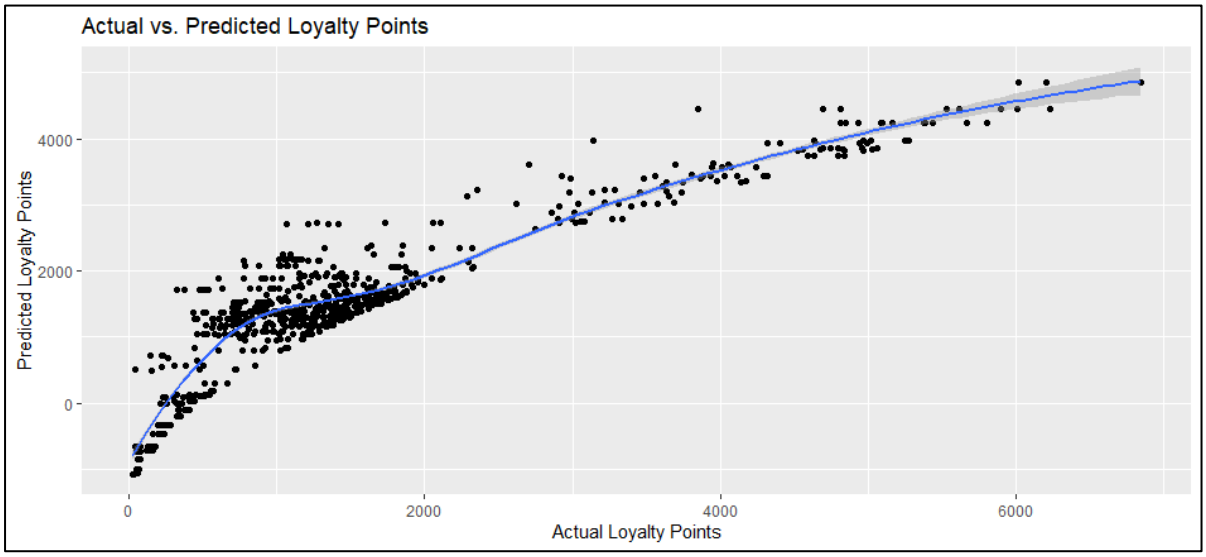






Q4: Can loyalty points be predicted?

The actual vs. predicted loyalty points visualisation shows that loyalty point accumulation could be modelled with a relatively high degree of accuracy.



Patterns and Predictions

Loyalty Point Drivers:

Remuneration and spending score were identified as the strongest predictors of loyalty point accumulation. The decision tree model indicated that these variables accounted for approximately 98% of model decision-making, whilst demographic characteristics contributed little predictive value. These findings suggest that Turtle Games should prioritise spend-based loyalty initiatives rather than demographic targeting.

Customer Segmentation:

Five customer segments were identified, three of which were commercially important: “moderate earners/ moderate spenders” (largest segment), “high earners/ high spenders” (highest-value segment), and “high earners/ low spenders” (greatest growth opportunity). Turtle Games should prioritise retaining the largest and highest-value segments whilst encouraging greater spending amongst high earners/ low spenders.

Customer Sentiment:

Most reviews were neutral to slightly positive in sentiment, with words such as “great”, “fun”, and “play” appearing frequently. This suggests that customers are generally satisfied with Turtle Games's products. However, instances of sentiment misclassification were identified, indicating that automated sentiment analysis should be supplemented by periodic manual review. Customer reviews should continue to be used to support product development and marketing activities.

Predictive Capability:

A multiple linear regression model built using remuneration and spending score explained approximately 83% of variation in loyalty points, demonstrating that loyalty point accumulation can be predicted with relatively high accuracy. This could support customer targeting, loyalty programme planning, and future marketing activities.

Business Recommendations

Priority 1: Target high earning, low spending customers:

Turtle Games should target “high earners/ low spenders” through tailored marketing campaigns and loyalty incentives, whilst continuing to retain “high earners/ high spenders” and “moderate earners/ moderate spenders”. Success could be measured through increased spending and loyalty point accumulation within these segments.

Priority 2: Improve the existing loyalty points programme:

Turtle Games should review its loyalty programme and introduce spend-based rewards aligned with customer spending behaviour. Success could be measured through increases in loyalty point balances across customer segments.

Priority 3: Enhance data collection:

Turtle Games should expand customer data collection to improve future predictive modelling. This could include surveying existing customers and collecting additional information during customer registration. Success could be measured through improvements on the 83% predictive accuracy achieved by the initial multiple linear regression model.

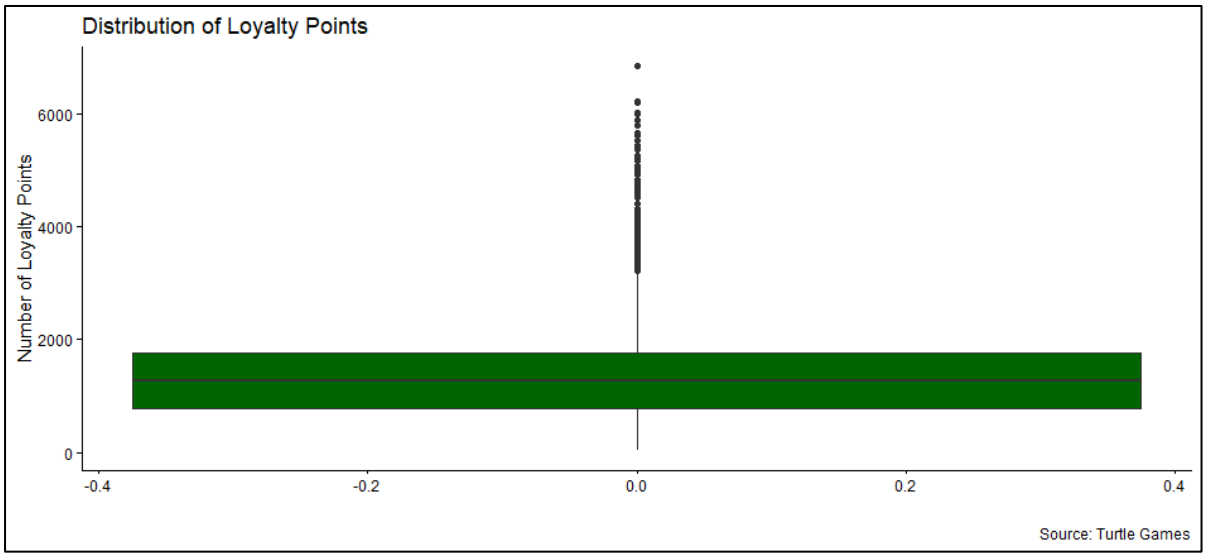
Technical Recommendations and Limitations

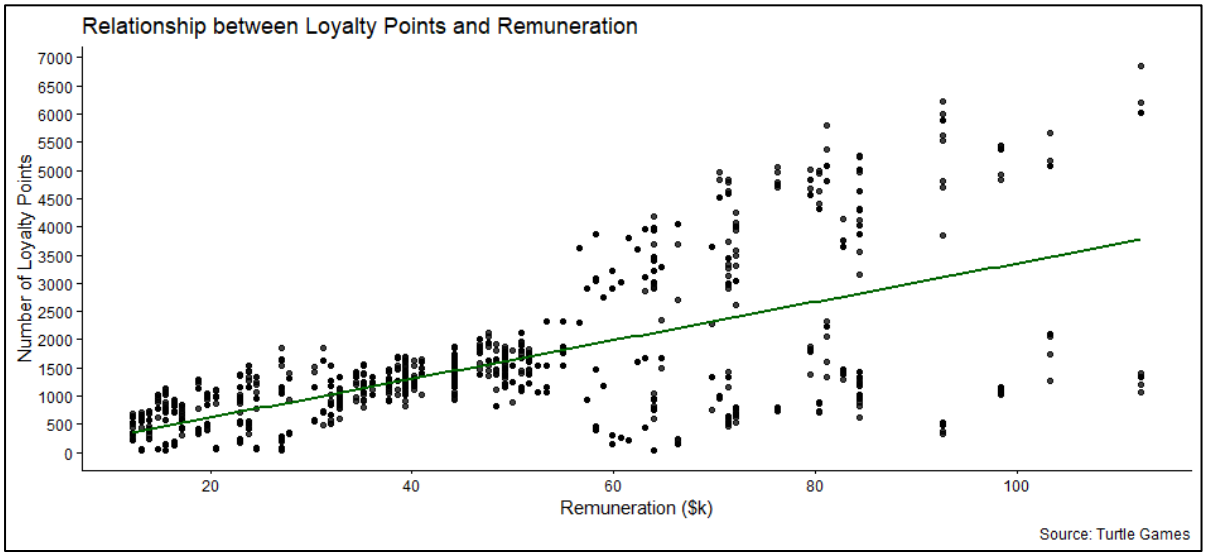
This analysis is limited to the information provided by Turtle Games. Consequently, some customer behaviours may be influenced by factors not included in the dataset. This is reflected in the multiple linear regression model, which explained approximately 83% of variation in loyalty points. The sentiment analysis model also occasionally misclassified reviews, indicating that automated sentiment analysis should be supplemented by periodic manual review.

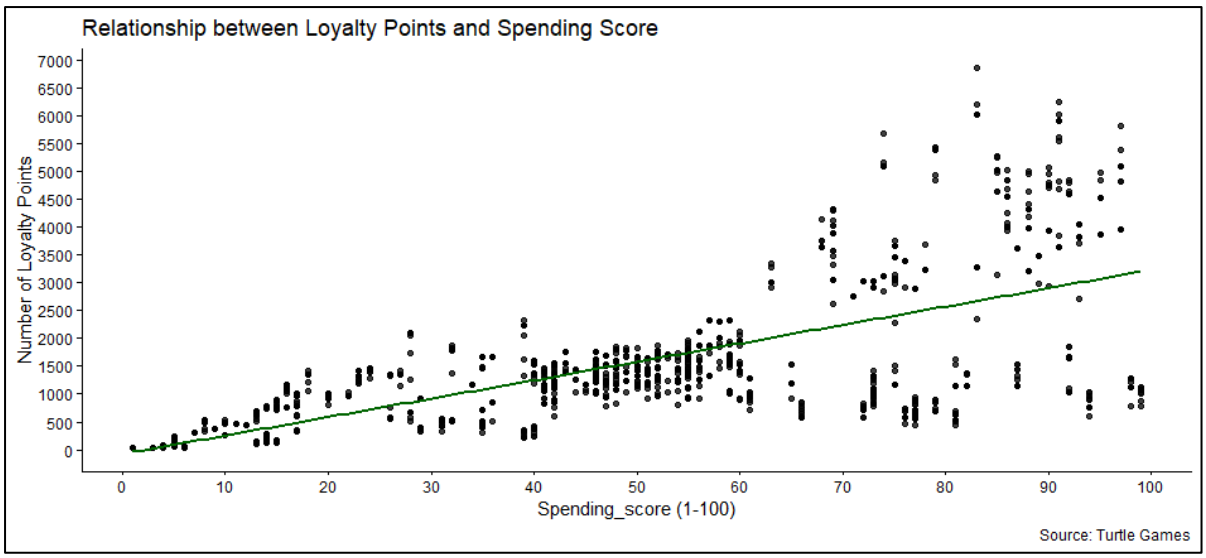
Future analyses could incorporate additional customer attributes, such as occupation and marital status, to improve predictive capability and explain variation in loyalty point accumulation not captured by the current models. Alternative predictive models could also be explored and compared with the multiple linear regression model. In addition, AI-powered natural language processing tools could be used to enhance sentiment analysis by identifying sentiment towards individual products rather than entire customer reviews or summaries.

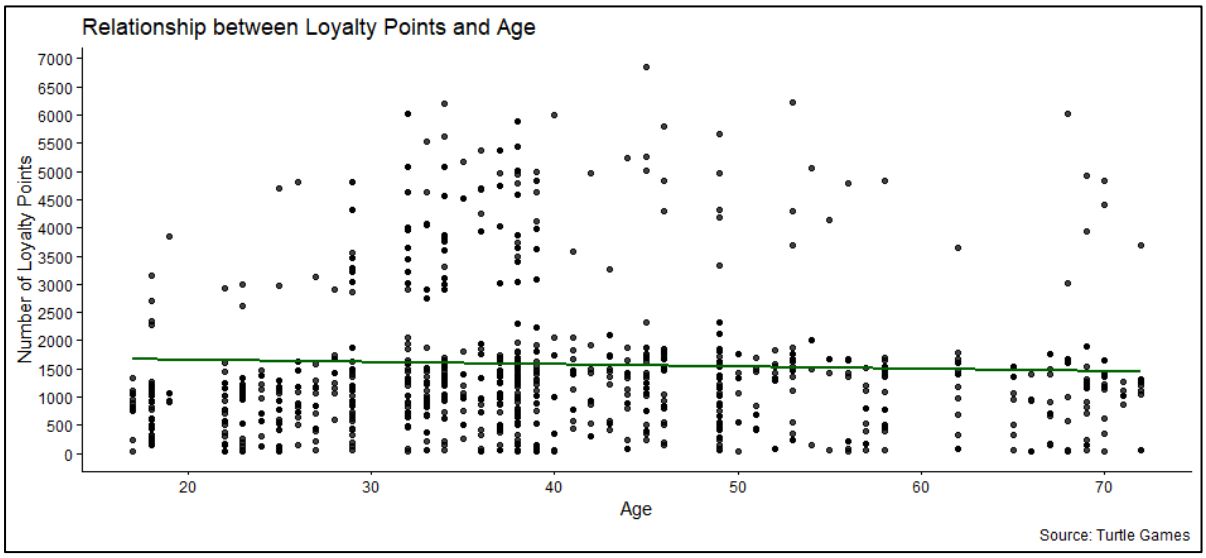
Appendices

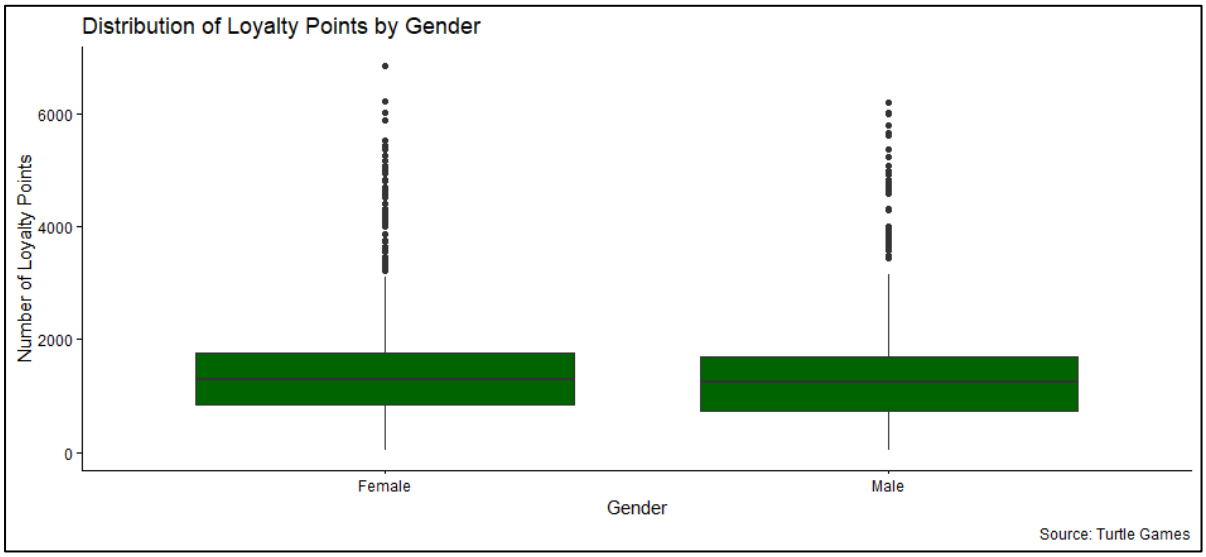
Appendix 1: Visualisations from Exploratory Data Analysis in R:

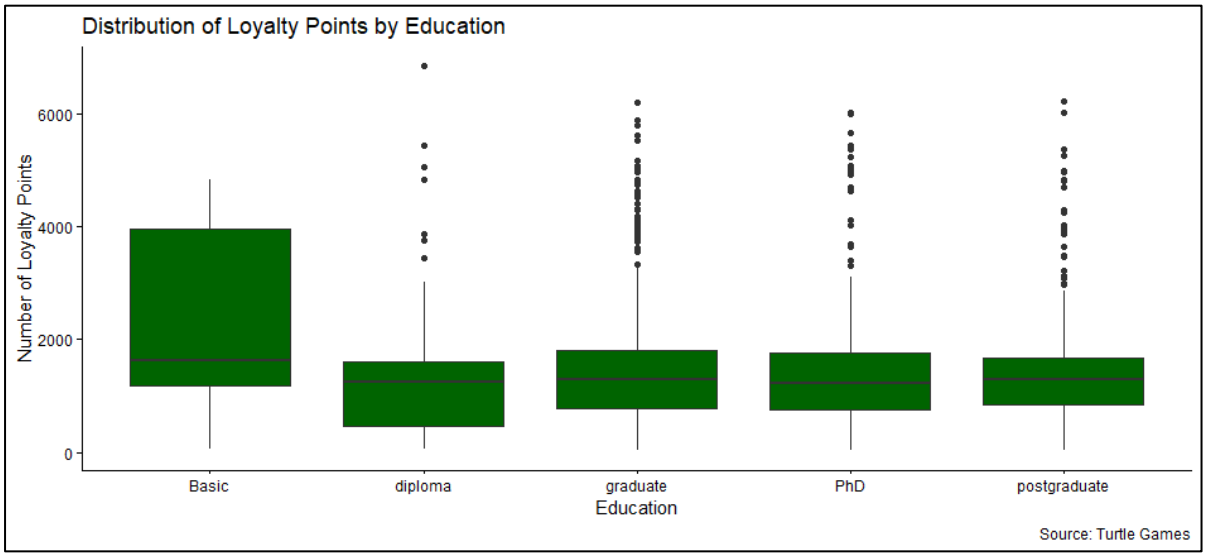


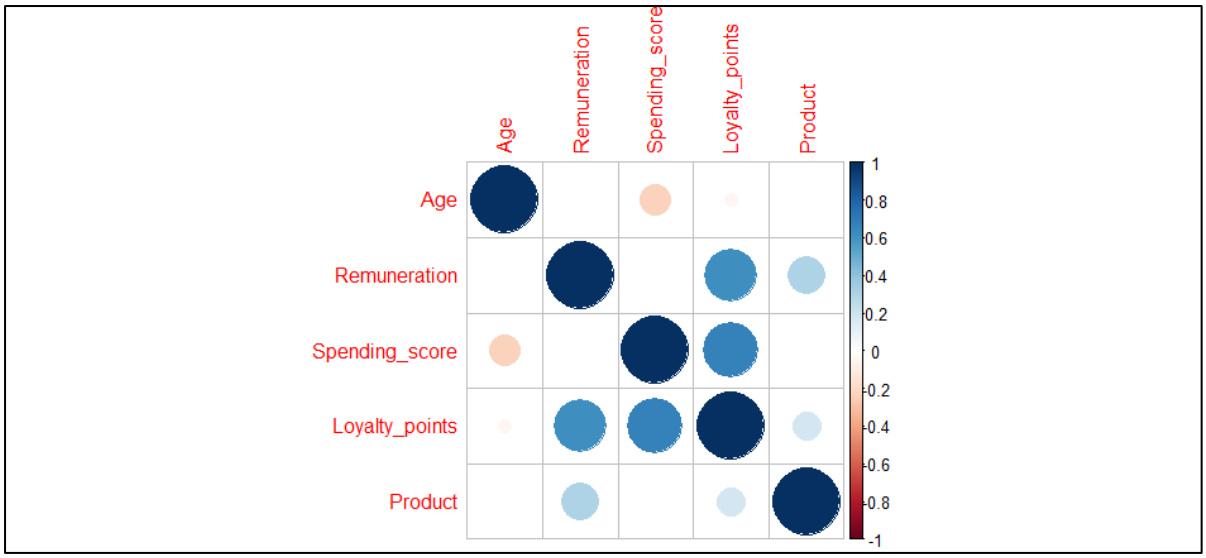



Relationship between Loyalty Points and Spending Score
Number of Loyalty Points
Spending_score (1-100)
Source: Turtle Games

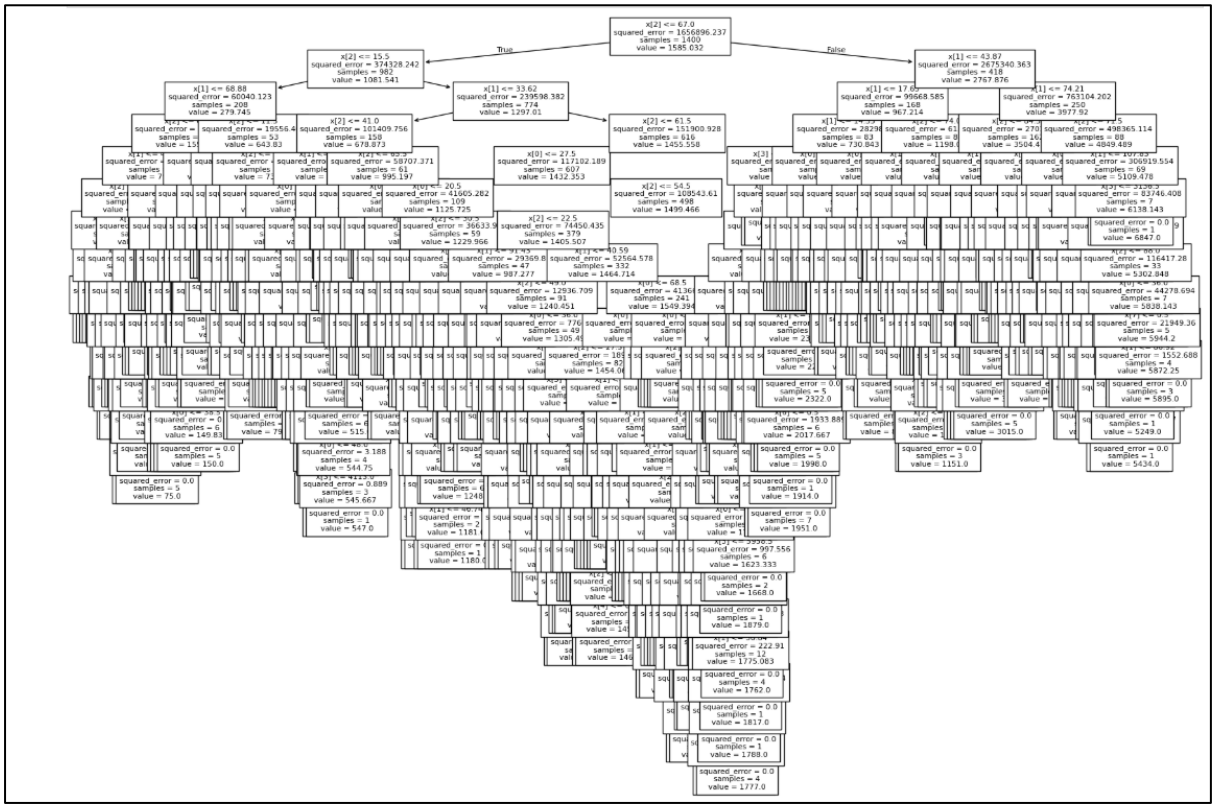

Relationship between Loyalty Points and Age
Number of Loyalty Points
Age
Source: Turtle Games


Distribution of Loyalty Points by Gender
Number of Loyalty Points
Female
Male
Gender
Source: Turtle Games

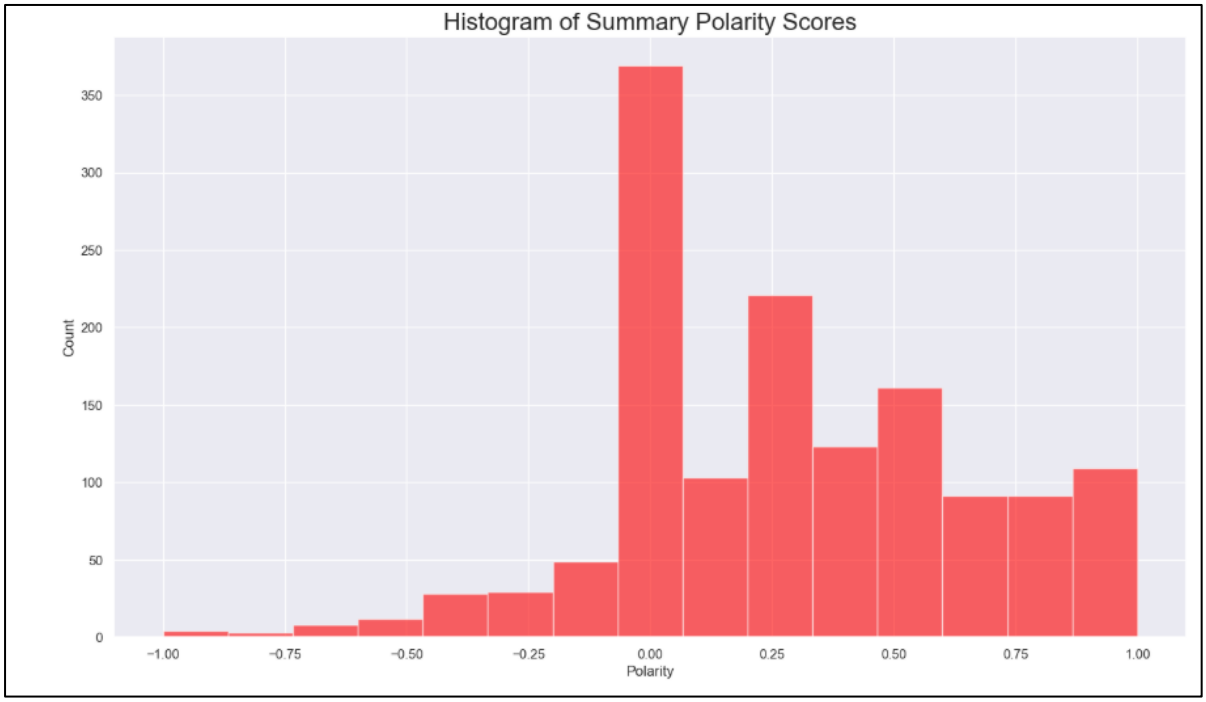

Distribution of Loyalty Points by Education
Number of Loyalty Points
6000
4000
2000
0
Basic
diploma
graduate
PhD
postgraduate
Education
Source: Turtle Games

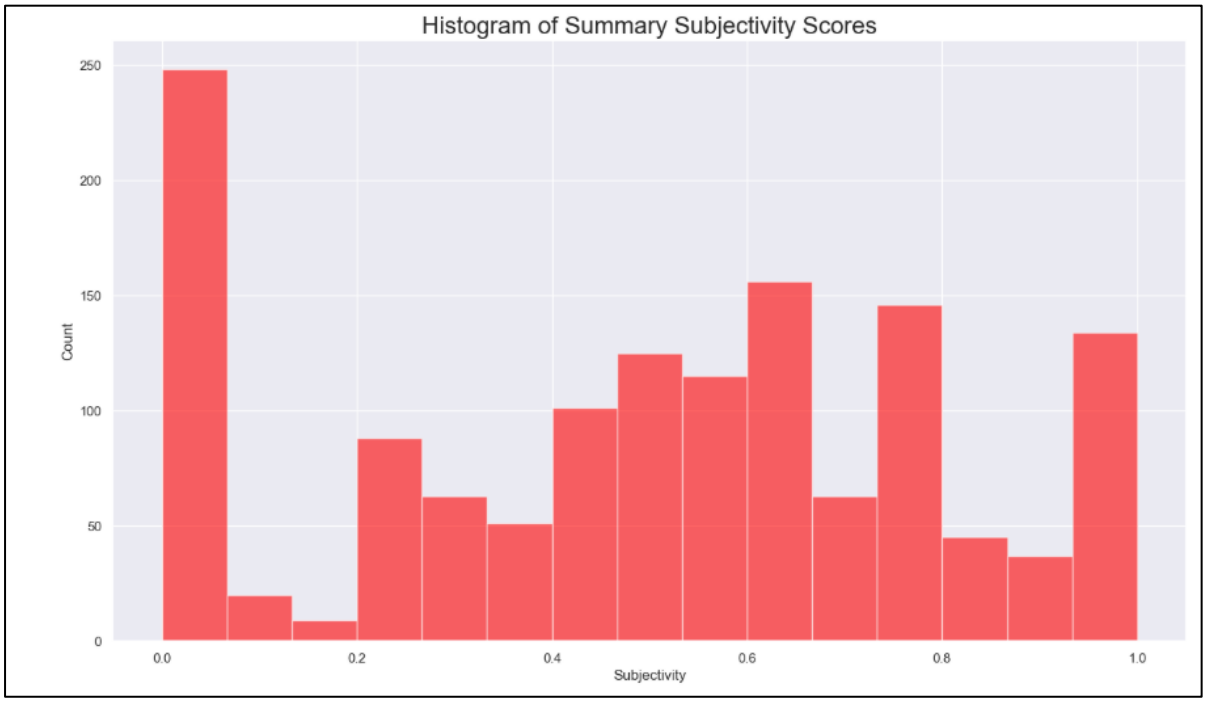

Age
Remuneration
Spending_score
Loyalty_points
Product
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1

Appendix 2: Decision tree visualisation:



Appendix 3: Additional sentiment analysis visualisations:




Histogram of Summary Subjectivity Scores
Count
Subjectivity
0
50
100
150
200
250
0.0
0.2
0.4
0.6
0.8
1.0